

Some Groups and Basic Properties

Road map

This chapter begins with the order of a group and three fundamental examples. It then develops Cayley tables, cancellation, translations, Abelian and cyclic groups, the complete classification of groups of order four, subgroups and the subgroup test, all subgroups of $\mathbb{Z}$, presentations by generators and relations, dihedral groups, and the quaternion group. Every computation and diagram in the handwritten lecture is reproduced below and supplemented with the intermediate steps needed by a first-time reader.

2.1 Order and first examples

We first recall the structure that is assumed throughout the lecture.

Definition 2.1: Group

A *group* is a set G equipped with a binary operation

$$G \times G \longrightarrow G, \quad (a, b) \longmapsto ab,$$

such that the following four axioms hold.

1. **Closure:** $ab \in G$ whenever $a, b \in G$.
2. **Associativity:** $(ab)c = a(bc)$ for all $a, b, c \in G$.
3. **Identity:** there is an element $e \in G$ such that $ea = ae = a$ for every $a \in G$.
4. **Inverses:** for each $a \in G$ there is an element $a^{-1} \in G$ such that $aa^{-1} = a^{-1}a = e$.

Definition 2.2: Order of a group

The *order* of G , denoted by $|G|$, is the number of elements in G . A finite group has $|G| \in \mathbb{N}$; for an infinite group one often writes $|G| = \infty$.

Example 2.1: The unique group of order one

If $|G| = 1$, then $G = \{e\}$. There is only one possible operation:

$$ee = e.$$

The identity is its own inverse, and

$$(ee)e = e = e(ee),$$

so associativity also holds. Thus this is a group, and it is the only group of order one.

Example 2.2: The additive group of integers

For $|G| = \infty$, a basic example is

$$G = (\mathbb{Z}, +).$$

For $a, b, c \in \mathbb{Z}$,

$$a + (b + c) = (a + b) + c, \quad a + 0 = 0 + a = a, \quad a + (-a) = (-a) + a = 0.$$

Thus 0 is the identity and $-a$ is the inverse of a . This is an infinite discrete group.

Example 2.3: The circle group $U(1)$

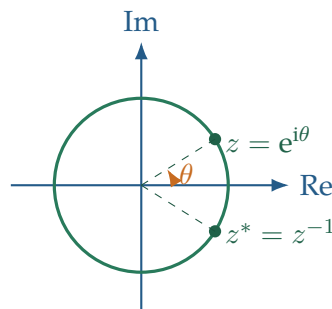
Another infinite group is

$$U(1) = \{z \in \mathbb{C} : |z| = 1\} = \{e^{i\theta} : \theta \in \mathbb{R}\},$$

with ordinary complex multiplication. Closure follows from $|zw| = |z||w| = 1$. Complex multiplication is associative, the identity is 1, and

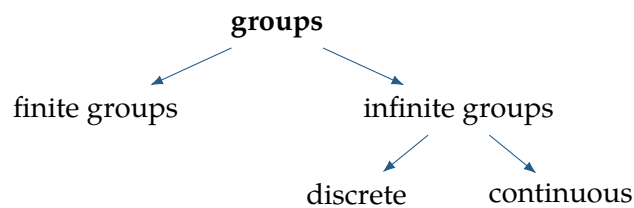
$$z^{-1} = \bar{z} = z^*$$

because $z\bar{z} = |z|^2 = 1$.



The inverse is therefore the reflection of z across the real axis, exactly as in the handwritten diagram.

The first broad classification in the notes is:



For example, finite groups are automatically discrete; $(\mathbb{Z}, +)$ is infinite and discrete; $U(1)$ is infinite and continuous.

2.2 Cayley tables, cancellation, and translations

2.2.1 Cayley tables

For a small finite group, the binary operation can be recorded completely in an operation table.

Definition 2.3: Cayley table

The *Cayley table* of a finite group explicitly specifies the map

$$G \times G \longrightarrow G, \quad (x, y) \longmapsto xy.$$

The entry in row x and column y is xy . Our convention is therefore “row element times column element.”

Example 2.4: The group of order two

Let $G = \{e, a\}$. The identity determines the first row and column:

$$\begin{array}{c|cc} \cdot & e & a \\ \hline e & e & a \\ a & a & ? \end{array}$$

We must have $a^2 = e$. Indeed, if $a^2 = a$, then

$$a^2 = a = ae,$$

and cancellation gives $a = e$, contradicting $|G| = 2$. Consequently

$$\begin{array}{c|cc} \cdot & e & a \\ \hline e & e & a \\ a & a & e \end{array} \quad \text{and} \quad G \cong \mathbb{Z}_2.$$

For $n \in \mathbb{N}$, the notation

$$a^n = \underbrace{a \cdot a \cdots a}_{n \text{ factors}}$$

will be used. We also set $a^0 = e$ and $a^{-n} = (a^{-1})^n$.

2.2.2 The cancellation rule

Lemma 2.1: Left and right cancellation

For $a, x, y \in G$,

$$ax = ay \implies x = y, \quad xa = ya \implies x = y.$$

More generally, each of the equations $ax = b$ and $xa = b$ has a unique solution:

$$ax = b \implies x = a^{-1}b, \quad xa = b \implies x = ba^{-1}.$$

证明. If $ax = ay$, multiply both sides on the left by a^{-1} :

$$a^{-1}(ax) = a^{-1}(ay),$$

$$(a^{-1}a)x = (a^{-1}a)y,$$

$$ex = ey,$$

$$x = y.$$

If $xa = ya$, multiply on the right by a^{-1} . The two displayed solutions are obtained in exactly the same way:

$$ax = b \Rightarrow a^{-1}ax = a^{-1}b \Rightarrow x = a^{-1}b,$$

$$xa = b \Rightarrow xaa^{-1} = ba^{-1} \Rightarrow x = ba^{-1}.$$

Uniqueness follows from cancellation. ■

Definition 2.4: Left and right translations

For fixed $a \in G$, define

$$\ell_a : G \longrightarrow G, \quad x \longmapsto ax, \quad r_a : G \longrightarrow G, \quad x \longmapsto xa.$$

These maps are called the *left translation* and *right translation* by a .

Proposition 2.1: Translations are bijective

For every $a \in G$, both ℓ_a and r_a are bijections.

证明. We prove the claim for ℓ_a .

Surjectivity. Pick $x \in G$ arbitrarily. Set

$$y = a^{-1}x.$$

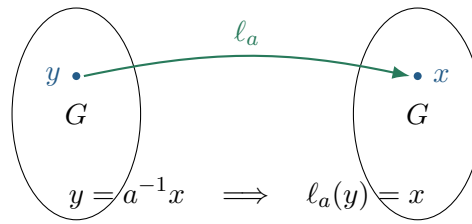
Then

$$\ell_a(y) = ay = a(a^{-1}x) = (aa^{-1})x = ex = x.$$

Thus every x has a preimage.

Injectivity. Suppose $\ell_a(x) = \ell_a(y)$. Then $ax = ay$, so left cancellation gives $x = y$. Therefore ℓ_a is one-to-one.

The proof for r_a is the same, using $y = xa^{-1}$ for surjectivity and right cancellation for injectivity. ■



The Sudoku rule for Cayley tables

The x -th row of a Cayley table lists the values of the bijection ℓ_x , while the y -th column lists the values of the bijection r_y . Hence every group element occurs exactly once in every row and exactly once in every column. This is the “Sudoku rule” used repeatedly in the handwritten notes.

Example 2.5: Translations in $\mathbb{Z}$ and $\mathbb{Z}_2$

In the additive group $\mathbb{Z}$,

$$\ell_k(m) = k + m = m + k = r_k(m).$$

Thus left and right translations coincide because addition is commutative. They shift the integer line by k :

$$\dots, -2, -1, 0, 1, 2, \dots \xrightarrow{\ell_k} \dots, k-2, k-1, k, k+1, k+2, \dots$$

In $G = \mathbb{Z}_2 = \{e, a\}$, the nontrivial translation interchanges the two elements:

$$\ell_a(e) = a, \quad \ell_a(a) = e.$$

For every group, $\ell_e = r_e = \text{id}_G$, because $ex = xe = x$.

2.3 Abelian and cyclic groups

Definition 2.5: Abelian group

A group G is *Abelian* if

$$ab = ba \quad \text{for all } a, b \in G.$$

The additive notation $a + b$ is often used for an Abelian group.

2.3.1 The unique group of order three

Proposition 2.2: There is essentially one group of order three

Every group of order three is cyclic and, after renaming its elements, is $\mathbb{Z}_3$.

证明. Write $G = \{e, a, b\}$. The identity first gives

$\cdot$	e	a	b
e	e	a	b
a	a	$?$	$?$
b	b	$?$	$?$

The entry a^2 cannot be a , by cancellation. It also cannot be e : if $a^2 = e$, the Sudoku rule forces $ab = b$, but $eb = b$, and right cancellation would imply $a = e$. Therefore $a^2 = b$. The row of a must then contain the remaining element e in its last position, so $ab = e$. The Sudoku rule now determines the final row:

$\cdot$	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

Since $b = a^2$ and $a^3 = e$, this is exactly the power table

$$G = \{e, a, a^2\} = \langle a \rangle.$$

Associativity is transparent from the power rule $(a^i a^j) a^k = a^{i+j+k} = a^i (a^j a^k)$, with exponents read modulo 3. ■

Although a table with three elements has $3^3 = 27$ individual associativity equalities

$$(xy)z = x(yz), \quad x, y, z \in G,$$

the power description proves all of them at once.

Definition 2.6: Cyclic group and generator

A group G is *cyclic* if there exists $a \in G$ such that

$$G = \langle a \rangle := \{a^k : k \in \mathbb{Z}\}.$$

The element a is then called a *generator* of G .

For all integers k, ℓ, m , the usual power rules hold:

$$\begin{aligned} (a^k a^\ell) a^m &= a^{k+\ell} a^m = a^{(k+\ell)+m} = a^{k+(\ell+m)} = a^k (a^\ell a^m), \\ a^0 a^k &= a^{0+k} = a^k, \\ a^{-k} a^k &= a^{-k+k} = a^0 = e, \\ a^k a^\ell &= a^{k+\ell} = a^{\ell+k} = a^\ell a^k. \end{aligned}$$

The last equality proves the next statement.

Proposition 2.3: Every cyclic group is Abelian

If $G = \langle a \rangle$, then G is Abelian.

证明. Given $x, y \in G$, choose $k, \ell \in \mathbb{Z}$ with $x = a^k$ and $y = a^\ell$. Then

$$xy = a^{k+\ell} = a^{\ell+k} = yx.$$

■

Example 2.6: $\mathbb{Z}$ and its generators

The additive group $\mathbb{Z}$ is cyclic:

$$\mathbb{Z} = \langle 1 \rangle.$$

In additive notation, the expression 1^n from multiplicative notation means a repeated sum:

$$\begin{cases} n = \underbrace{1 + \cdots + 1}_{n \text{ summands}}, & n \geq 1, \\ 0 = 1^0, & n = 0, \\ n = \underbrace{(-1) + \cdots + (-1)}_{|n| \text{ summands}}, & n < 0. \end{cases}$$

The element -1 is another generator. Indeed, for $n \geq 2$,

$$n = (-1)^{-n} = \underbrace{1 + \cdots + 1}_{n \text{ summands}}.$$

Thus a cyclic group need not have a unique generator.

Example 2.7: Two generators of $\mathbb{Z}_3$

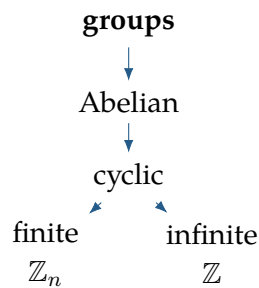
Let

$$G = \{a, a^2, a^3 = e\}.$$

Then a is a generator. The element a^2 is also a generator because

$$(a^2)^1 = a^2, \quad (a^2)^2 = a^4 = a, \quad (a^2)^3 = a^6 = e.$$

The hierarchy sketched in the notes is the following. The examples at the bottom are the cyclic groups; they are not intended to say that every Abelian group is cyclic.



2.4 Groups of order four

2.4.1 The cyclic possibility

Let $G = \{e, a, b, c\}$. One possible group is

$$\mathbb{Z}_4 = \{e, a, a^2 = b, a^3 = c\}, \quad a^4 = e,$$

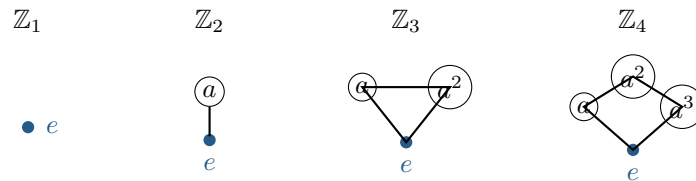
whose table is

$\cdot$	e	a	b	a^3
e	e	a	b	a^3
a	a	b	a^3	e
b	b	a^3	e	a
a^3	a^3	e	a	b

Equivalently, replacing the label a^3 by c ,

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

The cycle diagrams in the original notes give a graphical representation of the first four cyclic groups. An edge means “multiply by the displayed generator”; the filled vertex is the identity.



The diagrams immediately show that a^2 generates $\mathbb{Z}_3$ but does not generate $\mathbb{Z}_4$: in $\mathbb{Z}_4$, repeatedly multiplying by a^2 visits only e and a^2 .

2.4.2 Completing all order-four Cayley tables

The handwritten notes complete the table by a case split. We reproduce that argument explicitly. The identity row and column are fixed:

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	$\square$	$\square$	$\square$
b	b	$\square$	$\square$	$\square$
c	c	$\square$	$\square$	$\square$

The entry a^2 cannot equal a .

Case A: $a^2 = b$. The Sudoku rule forces

$$G_A : \begin{array}{c|cccc} \cdot & e & a & b & c \\ \hline e & e & a & b & c \\ a & a & b & c & e \\ b & b & c & e & a \\ c & c & e & a & b \end{array} .$$

Here a generates the whole group, so $G_A \cong \mathbb{Z}_4$.

Case B: $a^2 = e$, but $c^2 = a$. The second row and second column are first fixed by $a^2 = e$. One possible completion is

$$G_B : \begin{array}{c|cccc} \cdot & e & a & b & c \\ \hline e & e & a & b & c \\ a & a & e & c & b \\ b & b & c & a & e \\ c & c & b & e & a \end{array} .$$

Now c is a generator:

$$G_B = \{e, c, c^2 = a, c^3 = b\} \cong \mathbb{Z}_4.$$

Thus G_A and G_B are the same group after a relabelling of elements.

Case C: $a^2 = b^2 = c^2 = e$. The other possible completion is

$$K_4 : \begin{array}{c|cccc} \cdot & e & a & b & c \\ \hline e & e & a & b & c \\ a & a & e & c & b \\ b & b & c & e & a \\ c & c & b & a & e \end{array} .$$

This is the *Klein four group*.

Theorem 2.1: Classification of groups of order four

Up to a renaming that preserves multiplication, there are exactly two groups of order four:

$$\mathbb{Z}_4 = \langle a \mid a^4 = e \rangle, \quad K_4 = \{e, a, b, c\}, \quad a^2 = b^2 = c^2 = e.$$

证明. The preceding case split exhausts the possible values of a^2 and the remaining Sudoku completions. Cases A and B are cyclic and become identical after relabelling. Case C is not cyclic because every nonidentity element has order 2. Therefore it cannot be the same group as $\mathbb{Z}_4$. ■

What does “the same group” mean?

The handwritten notes ask how to define “essentially equal” or “isomorphic.” The formal definition appears later, but the idea is already visible here. A bijection $\varphi : G \rightarrow H$ is an isomorphism when

$$\varphi(xy) = \varphi(x)\varphi(y).$$

It changes labels but preserves the multiplication structure. For example, the isomorphism $G_A \rightarrow G_B$ can send $e \mapsto e$, $a \mapsto c$, $b = a^2 \mapsto a = c^2$, and $c = a^3 \mapsto b = c^3$.

2.4.3 Several models of K_4

The table of K_4 is symmetric across its main diagonal, so K_4 is Abelian. Associativity does hold; for example, it follows immediately from any of the concrete models below.

Example 2.8: Odd units modulo 8

The set

$$\{1, 3, 5, 7\}$$

is a group under multiplication modulo 8. Every element is its own inverse:

$$1^2 \equiv 3^2 \equiv 5^2 \equiv 7^2 \equiv 1 \pmod{8}.$$

Its multiplication table has exactly the pattern of K_4 . Hence

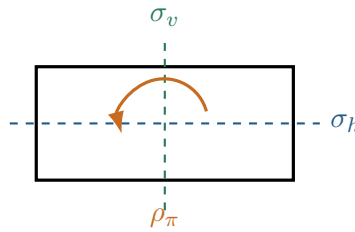
$$K_4 \cong \{1, 3, 5, 7\} \text{ under multiplication modulo 8.}$$

Example 2.9: The symmetry group of a rectangle

The symmetry group of a geometric object is the group of all transformations that leave the object invariant. A non-square rectangle has four such symmetries:

$$e, \quad \sigma_v \text{ (vertical reflection),} \quad \sigma_h \text{ (horizontal reflection),} \quad \rho_\pi \text{ (rotation by } 180^\circ\text{).}$$

Each nonidentity symmetry squares to e , and the product of any two different nonidentity symmetries is the third.



Thus the symmetry group of a rectangle is isomorphic to K_4 .

One may also write

$$K_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

The four pairs $(0, 0)$, $(1, 0)$, $(0, 1)$, $(1, 1)$, added componentwise modulo 2, again have precisely the same table.

2.5 Subgroups and the order of an element

The notes first point out that the highlighted 2×2 corner $\{e, a\}$ of the K_4 table is closed:

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Because the inherited operation keeps $\{e, a\}$ inside itself, and the remaining group axioms are inherited, $\{e, a\}$ is itself a group.

Definition 2.7: Subgroup

Let G be a group and $H \subseteq G$. We call H a *subgroup* of G , written $H \leq G$, if the operation of G , restricted to

$$H \times H \longrightarrow H,$$

makes H into a group.

The groups $\{e\}$ and G are always subgroups of G ; they are called the *trivial subgroups*. Associativity never needs to be re-proved for a subset: if $a, b, c \in H$, then $(ab)c = a(bc)$ because the same equality already holds in G . What must be checked is that the required products, identity, and inverses stay in H .

Lemma 2.2: One-step subgroup test

Let H be a nonempty subset of a group G . Then

$$H \leq G \iff ab^{-1} \in H \text{ for all } a, b \in H.$$

证明. *Necessity.* If $H \leq G$, then $b^{-1} \in H$ and closure gives $ab^{-1} \in H$.

Sufficiency. Assume $H \neq \emptyset$ and that $ab^{-1} \in H$ for every $a, b \in H$.

1. Choose $a \in H$. Taking the pair (a, a) gives

$$aa^{-1} = e \in H.$$

2. For any $a \in H$, apply the condition to e, a :

$$ea^{-1} = a^{-1} \in H.$$

3. If $a, b \in H$, then $b^{-1} \in H$ by the preceding step. Applying the condition to a, b^{-1} gives

$$a(b^{-1})^{-1} = ab \in H.$$

Thus H contains e , contains inverses, and is closed under products. Associativity is inherited from G , so $H \leq G$. ■

Proposition 2.4: The cyclic subgroup generated by an element

For every $a \in G$,

$$\langle a \rangle := \{a^k : k \in \mathbb{Z}\}$$

is a subgroup of G .

证明. The set is nonempty because $a^0 = e \in \langle a \rangle$. For $a^m, a^n \in \langle a \rangle$,

$$a^m(a^n)^{-1} = a^m a^{-n} = a^{m-n} \in \langle a \rangle.$$

The subgroup test completes the proof. ■

Definition 2.8: Order of an element

The *order* of $a \in G$ is

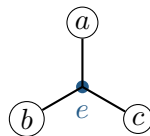
$$|a| := |\langle a \rangle|.$$

Equivalently, if some positive power of a is e , then $|a|$ is the smallest positive integer n with $a^n = e$. If no positive power equals e , then a has infinite order.

In K_4 , every nonidentity element has order 2:

$$\langle a \rangle = \{e, a\}, \quad \langle b \rangle = \{e, b\}, \quad \langle c \rangle = \{e, c\}.$$

The cycle graph drawn in the notes is therefore a three-armed star:



2.6 All subgroups of $\mathbb{Z}$

Definition 2.9: The subgroup $m\mathbb{Z}$

For $m \in \mathbb{N}$, define

$$m\mathbb{Z} := \{mk : k \in \mathbb{Z}\},$$

the set of all integer multiples of m . We also use $0\mathbb{Z} = \{0\}$.

The set $m\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$. Indeed, it is nonempty, and if $mk, m\ell \in m\mathbb{Z}$, then the additive form of xy^{-1} is a difference:

$$mk - (m\ell) = mk + m(-\ell) = m(k - \ell) \in m\mathbb{Z}.$$

The subgroup test applies.

Theorem 2.2: Subgroups of the integers

If $H \leq (\mathbb{Z}, +)$, then there is a unique $m \in \mathbb{N} \cup \{0\}$ such that

$$H = m\mathbb{Z}.$$

证明. If $H = \{0\}$, take $m = 0$. Now assume $H \neq \{0\}$. If $h \in H$ is nonzero, then either $h > 0$ or $-h > 0$, and $-h \in H$. Hence H contains a positive integer. By the well-ordering principle, there is a smallest positive integer $m \in H$.

Every integer multiple of m lies in H , so

$$m\mathbb{Z} \subseteq H.$$

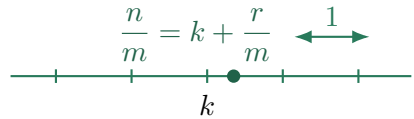
For the reverse inclusion, take any $n \in H$. By Euclidean division there are integers k, r such that

$$n = km + r, \quad 0 \leq r < m.$$

Equivalently,

$$\frac{n}{m} = k + \frac{r}{m}, \quad 0 \leq \frac{r}{m} < 1.$$

This is the decomposition illustrated by the number-line sketch:



Because $n, m \in H$, closure under integer multiples and subtraction gives

$$r = n - km \in H.$$

If $r > 0$, then r would be a positive element of H smaller than m , contradicting the choice of m . Thus $r = 0$, so $n = km \in m\mathbb{Z}$. Therefore $H \subseteq m\mathbb{Z}$, and hence $H = m\mathbb{Z}$.

For uniqueness, if $m\mathbb{Z} = n\mathbb{Z}$ with $m, n > 0$, then each is the smallest positive element of the same set, so $m = n$. The case $m = 0$ is also unique because $0\mathbb{Z} = \{0\}$. ■

2.7 Presentations: generators and relations

Definition 2.10: Presentation of a group

A presentation

$$G = \langle S \mid R \rangle$$

consists of a set S of *generators* and a set R of *relations* among those generators. Start with all finite words whose letters are the generators and their formal inverses, and then identify words using the relations in R .

Example 2.10: A presentation of K_4

The handwritten presentation is

$$K_4 = \langle a, b \mid a^2 = b^2 = (ab)^2 = e \rangle.$$

The relations imply $a^{-1} = a$ and $b^{-1} = b$. Moreover,

$$\begin{aligned} (ab)^2 = e &\implies abab = e, \\ &\implies bab = a^{-1} = a \quad (\text{multiply on the left by } a^{-1} = a), \\ &\implies ab = b^{-1}a = ba \quad (\text{multiply on the left by } b^{-1} = b). \end{aligned}$$

Thus a and b commute. Any word can now be rearranged into the form $a^i b^j$, and the squares can be deleted, leaving $i, j \in \{0, 1\}$. Consequently every word represents one of

$$e, \quad a, \quad b, \quad ab.$$

For example,

$$ba = ab, \quad a^2 = e, \quad b^2 a = a, \quad a^{-1} b^{-1} = ab.$$

These are precisely the four elements of K_4 .

Associativity does not have to be added as another relation: multiplication of words is concatenation, and concatenation is associative. Formally, a presentation constructs a quotient of a free group, so the group axioms are automatically satisfied.

Example 2.11: A presentation of $\mathbb{Z}_n$

$$\mathbb{Z}_n = \langle a \mid a^n = e \rangle = \{e, a, a^2, \dots, a^{n-1}\}.$$

Any higher or negative power is reduced by changing the exponent modulo n . For instance $a^{n+2} = a^2$ and $a^{-1} = a^{n-1}$.

What presentations do and do not make easy

- One group may have many different presentations.
- A presentation can define a group very compactly.
- Important properties, even the order of the group, may not be obvious from a presentation.
- Every group has a presentation: in a deliberately redundant one, use every group element as a generator and the entire Cayley table as the relations.

Definition 2.11: Free group

For a set $S = \{a, b, \dots\}$, the *free group on S* is

$$F(S) = \langle S \mid \emptyset \rangle.$$

Its elements are reduced words in the letters s and s^{-1} ($s \in S$), with only adjacent inverse pairs ss^{-1} and $s^{-1}s$ cancelled.

If S contains at least two generators, the free group is non-Abelian by construction: the words ab and ba are distinct because no relation permits one to be changed into the other.

2.8 The dihedral groups

2.8.1 The group D_3

Example 2.12: A six-element presentation

Consider

$$G = \langle a, b \mid a^3 = e, b^2 = e, bab^{-1} = a^2 \rangle.$$

Since $b^{-1} = b$ and $a^2 = a^{-1}$, the third relation is equivalently

$$ba = a^2b.$$

This rule moves every b to the right of every power of a . After using $a^3 = e$ and $b^2 = e$, every word reduces to one of

$$G = \{e, a, a^2, b, ab, a^2b\}.$$

The Cayley table is

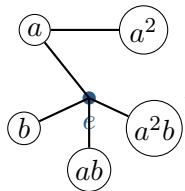
$\cdot$	e	a	a^2	b	ab	a^2b
e	e	a	a^2	b	ab	a^2b
a	a	a^2	e	ab	a^2b	b
a^2	a^2	e	a	a^2b	b	ab
b	b	a^2b	ab	e	a^2	a
ab	ab	b	a^2b	a	e	a^2
a^2b	a^2b	ab	b	a^2	a	e

The table is not symmetric; for example,

$$ab \neq ba = a^2b.$$

Therefore G is non-Abelian.

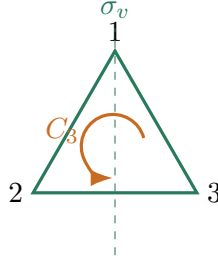
The cycle graph in the notes makes all cyclic subgroups visible. The rotation subgroup $\{e, a, a^2\}$ is a triangle, while b, ab, a^2b each generate a two-element subgroup.



Definition 2.12: Dihedral group

The *dihedral group* D_n is the symmetry group of a regular polygon with n sides: the group of all transformations that leave the polygon invariant. The integer n is called the degree.

For a regular triangle, let C_3 be counterclockwise rotation through $2\pi/3$, and let σ_v be reflection in the vertical axis.



As permutations of the three vertices,

$$C_3 = (123), \quad \sigma_v = (23).$$

Because $\sigma_v^{-1} = \sigma_v$,

$$\sigma_v C_3 \sigma_v^{-1} = (23)(123)(23) = (132) = C_3^2 = C_3^{-1}.$$

Equivalently,

$$\sigma_v C_3 = C_3^2 \sigma_v.$$

Thus

$$D_3 = \langle C_3, \sigma_v \mid C_3^3 = \sigma_v^2 = e, \sigma_v C_3 \sigma_v^{-1} = C_3^2 \rangle.$$

Identifying $a = C_3$ and $b = \sigma_v$ recovers the six-element group and the Cayley table above.

The smallest non-Abelian group

D_3 is the smallest non-Abelian group. Groups of orders 1, 2, 3 and 4 are Abelian by the classifications already discussed; a group of prime order 5 is cyclic (proved later using Lagrange's theorem). At order 6, D_3 provides the first non-Abelian example.

2.8.2 The general group D_n

Let C_n be rotation by $2\pi/n$, and let σ be a fixed reflection. Then

$$D_n = \langle C_n, \sigma \mid C_n^n = e, \sigma^2 = e, \sigma C_n \sigma^{-1} = C_n^{-1} \rangle.$$

The relation says that reflecting, rotating, and reflecting back reverses the direction of rotation. Every word can be reduced by moving σ to the right:

$$\sigma C_n = C_n^{-1} \sigma.$$

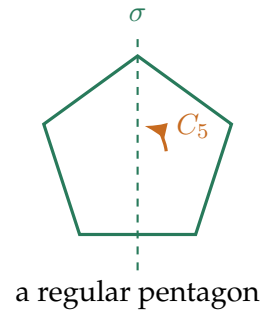
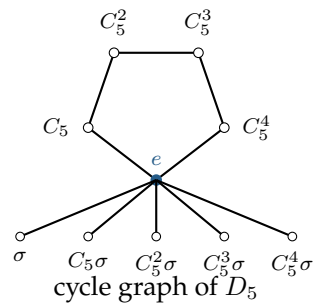
After also reducing powers modulo n , the elements are

$$D_n = \{e, C_n, C_n^2, \dots, C_n^{n-1}, \sigma, C_n \sigma, C_n^2 \sigma, \dots, C_n^{n-1} \sigma\}.$$

There are n rotations and n reflections, so

$$|D_n| = 2n.$$

For $n = 5$, the two sketches from the notes may be drawn together:



The small-order list recorded in the notes is:

$ G $	2	3	4	5	6	7
groups listed	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_4, K_4$	$\mathbb{Z}_5$	$\mathbb{Z}_6, D_3$	$\mathbb{Z}_7$

For orders 2, 3, 5, 7, the cyclic group is the only possibility. At order 4 there are two groups; at order 6 the list already contains both an Abelian and a non-Abelian group. Starting at order 8, further kinds of groups appear.

2.9 The quaternion group Q_8

Example 2.13: Presentation and multiplication rules

The quaternion group has the presentation

$$Q_8 = \langle a, b \mid a^4 = e, b^2 = a^2, bab^{-1} = a^{-1} \rangle.$$

An equivalent and more familiar notation is

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\},$$

with

$$i^2 = j^2 = k^2 = ijk = -1, \quad (-1)^2 = 1.$$

Here 1 is the identity and -1 is a central element of order 2. The six imaginary units occur in inverse pairs:

$$-i = (-1)i = i^{-1}, \quad -j = j^{-1}, \quad -k = k^{-1}.$$

The fundamental products are

$$ij = k, \quad jk = i, \quad ki = j,$$

whereas reversing the order changes the sign:

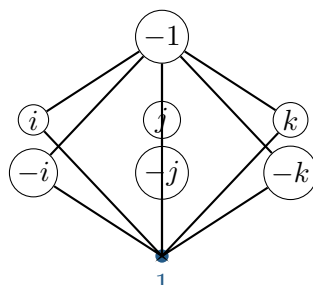
$$ji = -k, \quad kj = -i, \quad ik = -j.$$

In particular,

$$ij = k \neq -k = ji,$$

so Q_8 is non-Abelian and $|Q_8| = 8$.

The original cycle graph consists of three cyclic subgroups of order 4 sharing the vertices 1 and -1 :



each diamond is a four-element cyclic subgroup

The three diamonds are $\langle i \rangle$, $\langle j \rangle$, and $\langle k \rangle$. Their common top and bottom vertices are the common elements -1 and 1 .

The quaternion algebra is not a field

The symbols $1, i, j, k$ span the real quaternion algebra

$$\mathbb{H} = \{a + ib + jc + kd : a, b, c, d \in \mathbb{R}\}.$$

Although every nonzero quaternion has a multiplicative inverse, multiplication is not commutative; for example $ij \neq ji$. Under the standard definition, a field must have commutative multiplication, so $\mathbb{H}$ is a noncommutative division algebra rather than a field.

The lecture closes with the scale of finite-group classification: groups of order at most 2000 have been tabulated, and the note records

49,910,529,484

different isomorphism classes in that range. Even though the first few orders have very short lists, the number of possible group structures grows dramatically.

What to retain from this chapter

1. Cancellation makes every left and right translation a bijection, so every Cayley-table row and column is a permutation of the group.
2. Cyclic groups are Abelian, but an Abelian group need not be cyclic; K_4 is the first example in these notes.
3. The only groups of order four are $\mathbb{Z}_4$ and K_4 .
4. A nonempty subset $H \subseteq G$ is a subgroup exactly when $ab^{-1} \in H$ for all $a, b \in H$.
5. Every subgroup of $(\mathbb{Z}, +)$ is $m\mathbb{Z}$.
6. Presentations turn generators and relations into groups. They give compact descriptions of K_4 , D_n , and Q_8 .
7. D_3 is the first non-Abelian group, while Q_8 illustrates a genuinely new group structure at order eight.